FYS5429/9429 March 12

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FFNN

$$x \rightarrow \boxed{\text{Encoder}} - \boxed{h} - \boxed{\text{Decoder}} \rightarrow \hat{x}$$

FFNN

Latent degrees of freedom

$$h = f(x, W_e) \wedge \hat{x} = g(h, W_d)$$

Reconstruction error

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (x_i' - \hat{x}_i)^2$$

$$= \frac{1}{n} \sum_{i=1}^{n} \left(x_i' - g(W_D, f(x_i' W_e))\right)^2$$

$$\underset{W_D, W_e}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \left(\qquad \right)^2$$

in general x_i' etc are matrices

optimize $\|X - \hat{X}\|_F^2$

$$= \mathrm{Tr}\left((X-\hat{X})^T(X-\hat{X})\right)$$

$$\|A\|_F^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2 = \mathrm{Tr}(A^TA)$$

$(A \in \mathbb{R}^{m \times n})$

$$(A^TA)_{jk} = \sum_{i=1}^{m} a_{ij}\, a_{ik}$$

$$\mathrm{Tr}(A^TA) = \sum_{j=1}^{n} (A^TA)_{jj}$$

$$\mathrm{Tr}(A^T A) = \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij}^2$$

$$= \|A\|_F^2$$

Principal comp analysis

and AES

$$\hat{x} = g(W_d, f(x, W_e))$$

Linear connection !

$$h = f(x, W_e) = W_e x$$

$$\hat{x} = g(h, W_d) = W_d \cdot h$$

$$= W_d W_e x$$

$$\underset{W_d\, W_e}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} \left(x_i - W_d W_e x_i\right)^2$$

Proof of link between linear Autoencoder and PCA.

input data $X \in \mathbb{R}^{n \times p}$

output — " — $y \in \mathbb{R}^{n}$

$$X = \begin{bmatrix} x_{00} & x_{01} & \cdots & x_{0\,p-1} \\ x_{10} & & & \\ \vdots & & & \\ x_{n-1\,0} & \cdots & \cdots & x_{n-1\,p-1} \end{bmatrix}$$

Data entries

Features

Model $\tilde{y} = X \cdot \theta$

$$\hat{\theta} = \frac{1}{X^T X} X^T y$$

(Ordinary least squares)

$$\tilde{y} = \underbrace{X \frac{1}{X^T X} X^T}_{A} y$$

$$A^2 = A$$

$$\tilde{y} = A \cdot y$$

$$W_d W_e = A^2 = A$$

no dimension reduction.

$$\frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{n} \left((y - X\hat{\theta})^T (y - X\theta) \right)$$

<u>SVD</u>

$$X = UDV^T$$

$$U \in \mathbb{R}^{n \times n} \qquad U^T U = U U^T = \mathbb{1}$$

$$V \in \mathbb{R}^{p \times p} \qquad V^T V = V V^T = \mathbb{1}$$

$$D = \begin{bmatrix} \lambda_0 & & & & 0 \\ & \lambda_1 & & & \\ & & \ddots & & \\ & & & \lambda_{p-1} & \\ 0 & & & & 0 \end{bmatrix}$$

singular value

$$\lambda_0 > \lambda_1 > \lambda_2 \ldots > \lambda_{p-1} > 0 \ldots 0$$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (3 \times 2)$$

$$U = \begin{bmatrix} \vec{u}_0 & \vec{u}_1 & \cdots & \vec{u}_{n-1} \end{bmatrix}$$

$$V = \begin{bmatrix} \vec{v}_0 & \vec{v}_1 & \cdots & \vec{v}_{p-1} \end{bmatrix}$$

$$\vec{u}_i^T \vec{u}_j = \delta_{ij}$$

$$\vec{v}_i^T \vec{v}_j = \delta_{ij}$$

OLS

$$\tilde{y} = X \frac{1}{X^T X} X^T y$$

$$= U D V^T \frac{1}{V D^T \underbrace{U^T U}_{\mathbb{1}} D V^T} V D^T U^T y$$

A and B, square and invertible

$$\frac{1}{AB} = B^{-1} A^{-1}$$

$$\frac{1}{V \underbrace{D^T D} V^T}$$

$$D^T D = \begin{bmatrix} \lambda_0^2 & & & \\ & \ddots & & \\ & & \lambda_{p-1}^2 \end{bmatrix}$$

$$\underbrace{V}_{A} \underbrace{D^T D V^T}_{B}$$

$$\left(D^T D V^T\right)^{-1} V^{-1}$$

$$\tilde{y} = U D V^T \cdot V \frac{1}{D^T D} V^T V D^T U^T y$$

$I_{p \times p}$ $\quad$ $p \times p$ $\quad$ $I_{p \times p}$

$n \times n$ $\quad$ $n \times p$ $\quad$ $p \times n$ $\quad$ $n \times n$ $\quad$ n

$$\tilde{y} = \left(\sum_{i=0}^{p-1} u_i u_i^T \right) y$$

PCA means that we want an effective dim

$m \leq p$

$$W_d W_E y \cong \left(\sum_{i=0}^{m} u_i u_i^T\right) y$$

only one vector

$$u_0 u_0^T = W_d W_e$$

Reconstruction error $\propto \|(y - u_0 u_0^T y)\|_2^2$

What is PCA?

$$X \in \mathbb{R}^{n \times p}$$

$$X = \left[\vec{x}_0 \; \vec{x}_1 \; \cdots \; \vec{x}_{p-1} \right]$$

$$\vec{x}_i = \begin{bmatrix} x_{0i} \\ x_{1i} \\ \vdots \\ x_{n-1\,i} \end{bmatrix} \qquad \bar{x}_i = \frac{1}{n} \sum_k x_{ki}$$

$$\mathrm{cov}[\vec{x}_i \vec{x}_j] = \frac{1}{m} \sum_k (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$\tilde{x}_{ki} = x_{ki} - \bar{x}_i \rightarrow x_{ki}$$

centered (from scaling)

$$X = [\vec{x}_0 \; \vec{x}_1] = \begin{bmatrix} x_{00} & x_{01} \\ x_{10} & x_{11} \\ x_{20} & x_{21} \end{bmatrix}$$

$$\mathrm{cov}[\vec{x}_0 \vec{x}_1] = \frac{1}{3}(x_{00}x_{01} + x_{10}x_{11} + x_{30}x_{21})$$

$$= \underbrace{\begin{bmatrix} x_{00} & x_{10} & x_{20} \end{bmatrix}}_{\vec{x}_0^T} \underbrace{\begin{bmatrix} x_{01} \\ x_{11} \\ x_{21} \end{bmatrix}}_{\vec{x}_1}$$

$$\Rightarrow \text{cov}[\vec{x}_0 \, \vec{x}_1] = \vec{x}_0^T \vec{x}_1 / 3$$

Covariance matrix

$$\text{cov}[X] = \frac{1}{n} X^T X$$

$$(\vec{x}_0, \vec{x}_1)$$

$$\mathrm{cov}[X] = \frac{1}{n}\begin{bmatrix} \vec{x}_0^T \vec{x}_0 & \vec{x}_0^T \vec{x}_1 \\ \vec{x}_1^T \vec{x}_0 & \vec{x}_1^T x \end{bmatrix}$$

$$\frac{\vec{x}_i^T \vec{x}_i}{n} = \frac{1}{n}\sum_k x_{ki}^2 = \sigma_i^2$$

$$\left(\frac{1}{n}\sum_k (x_{ki} - \bar{x}_i)^2\right)$$

$$\mathrm{cov}[X] = \frac{1}{n} X^T X$$

SVD again

$$\mathrm{cov}[X] = \frac{1}{n} V D^T \underbrace{U^T U}_{\mathbb{1}} D V^T$$

$$= \frac{1}{n} V D^T D V^T$$

multiply with V from the right

$$\mathrm{cov}[X]\, V = \frac{1}{n} V \underbrace{D^T D}_{} \underbrace{V^T V}_{\mathbb{1}}$$

$$\begin{bmatrix} \lambda_0^2 & & \\ & \ddots & \\ & & \lambda_{p-1}^2 \end{bmatrix}$$

$$V = \begin{bmatrix} \vec{v}_0 & \vec{v}_1 & \cdots & \vec{v}_{p-1} \end{bmatrix}$$

$$\mathrm{cov}[X]\, \vec{v}_i = \lambda_i^2 v_i$$

$$\frac{\lambda_i^2}{n} = \sigma_i^2 \quad \text{(variance)}$$

assume $S^T S = S S^T = \mathbb{1}$

$$S^T \operatorname{cov}[x] S = \operatorname{cov}[y]$$

$$= \begin{bmatrix} \sigma_0^2 & & \\ & \ddots & \\ & & \sigma_{p-1}^2 \end{bmatrix}$$

$$S = [\vec{s}_0 \cdots \vec{s}_{p-1}] \quad \vec{s}_i^T \vec{s}_j = \delta_{ij}$$

PCA set a cutoff on total variance

$$\sum_{i=0}^{p-1} \sigma_i^2 = \sigma^2_{Total}$$

$$m \leq p$$

$$\sum_{i=0}^{m-1} \sigma_i^2 \leq \text{cutoff}$$

PCA – theorem

$$\tilde{y} = \vec{s}_0 \vec{s}_0^T y \qquad \vec{s}_0^T s_0 = 1$$

Reconstruction error

$$\| (y - \underbrace{\vec{s}_0 \vec{s}_0^T}_{W_d W_e} y) \|_2^2 \leq \varepsilon$$